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Auto-eD: A visual learning tool for automatic differentiation

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Software

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Summary

Most fields of scientific inquiry require the evaluation of derivatives to calculate and optimize quantities of interest. Automatic differentiation is a set of techniques that allow the differentiation of computer programs to machine precision without requiring full symbolic derivatives (Baydin et al., 2017; Griewank, 1989). The great success of machine learning algorithms, and neural networks in particular, was partly enabled by the celebrated back-propagation algorithm (Werbos, 1990), which is a special case of automatic differentiation. Given the rapidly increasing interest in algorithms that rely on automatic differentiation, and the evolution towards differential programming paradigms (Innes et al., 2019), it is important that students be taught the basics of this key family of algorithms.

Automatic differentiation is a method of computing derivatives to machine precision based on the decomposition of functions into a series of elementary operations. These operations can be conceptualized as forming a graph structure. This graph can be traversed in the forward or reverse direction, giving rise to the two primary modes in automatic differentiation. The goal of the `Auto-eD` software and the accompanying lecture modules is to enhance students' understanding of automatic differentiation by helping them to visualize the underlying graph structure of the computations.

Statement of Need

While most students encountering automatic differentiation will be familiar with the chain rule from multivariable calculus, far fewer students understand how to relate the chain rule to methods for computing derivatives in software. The fundamental step to this understanding is the ability to decompose a function into elementary operations and traverse the resulting graph structure. The `Auto-eD` software and lecture modules provide a framework for educators to help teach students how to relate functions to an underlying graph and use that graph to compute derivatives. Using `Auto-eD`, students are able to visualize the forward and reverse graph and an accompanying computational trace table. As a result, students gain a better understanding of the process of automatic differentiation and hence are better equipped to understand its use in a wide range of applications.

Content

The content available in the Auto-eD package contains a software package capable of performing automatic differentiation for a function and visualizing this calculation in a

table and graphs. Additionally, the `Auto-eD` package content contains a learning unit for teaching automatic differentiation through an easily-accessible web application based on this software.

Auto-eD Visualization Software

The `Auto-eD` package can be accessed using two different methods. For students more familiar with Python and coding, the code available in the modules `ADnum.py`, `ADmath.py`, and `ADgraph.py` allows a user to perform automatic differentiation while visualizing the underlying graphs and computations. These modules provide the functionality to visualize the graphs underlying forward and reverse mode. An experienced user can write a Python script that imports these modules and provides the capability to dynamically visualize the traceback of reverse mode through the graph. The resulting graphs and tables can be resized for enhanced interactivity. The process is outlined in `DeveloperDocumentation.ipynb`, which can be found at the top level of the Auto-eD package.

This software and associated web application are valuable pedagogical tools because they allow students to view the computational graph in both forward and reverse mode alongside the computational table. This makes it easy for students to relate the table and the graph as well as compare the differences in graph traversal of the forward and reverse mode. An example output from the web application is below. The first row shows the computational graphs for the forward (left) and reverse (right) modes of automatic differentiation. The table in the second row shows the values of the function and it's derivative at a specified evaluation point at each step of the graph. The figure in the bottom right shows a snapshot of an interactive visualization that enables the user to systematically step through each step of reverse mode. The current step is highlighted with a bold, yellow arrow.

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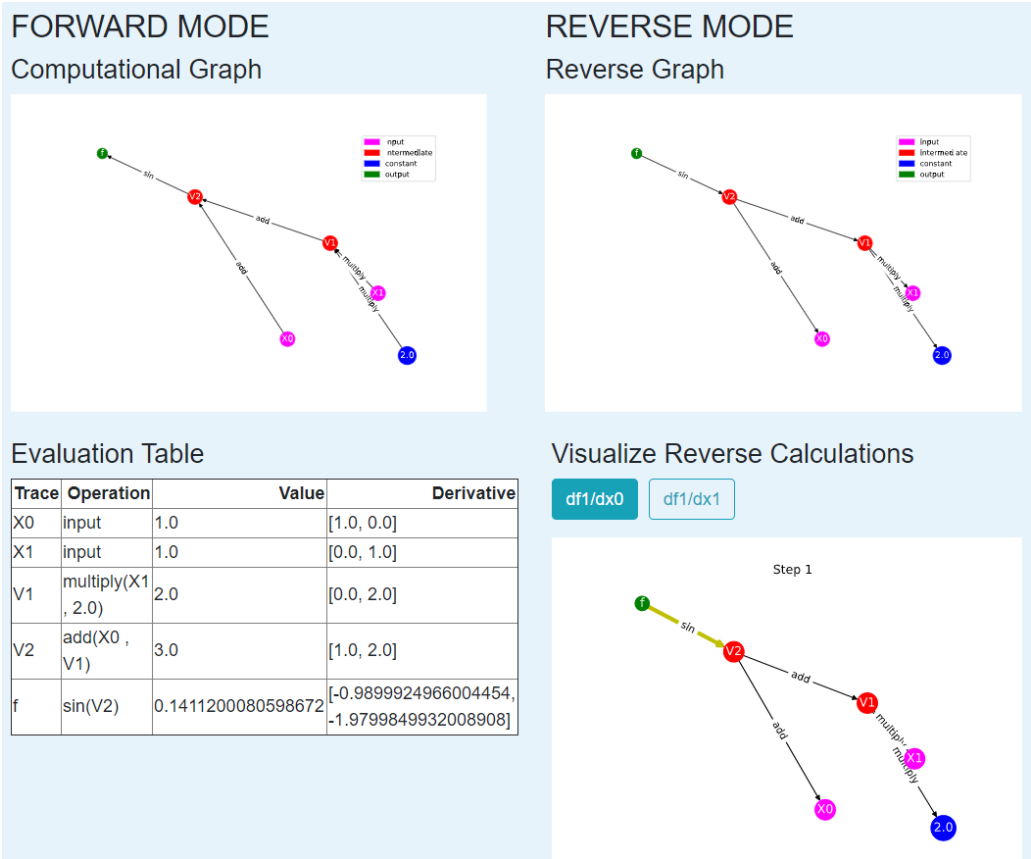

Trace	Operation	Value	Derivative
X0	input	1.0	[1.0, 0.0]
X1	input	1.0	[0.0, 1.0]
V1	multiply(X1 , 2.0)	2.0	[0.0, 2.0]
V2	add(X0 , V1)	3.0	[1.0, 2.0]
f	sin(V2)	0.1411200080598672	[-0.9899924966004454, -1.9799849932008908]

Figure 1: demo

For ease of instructional use for students less familiar with Python and coding, `Auto-eD` is available as a web application. If the web application is slow to load, it can alternatively be run locally by downloading the code from Github and launching `ADapp.py` from the command line. To make sure this final option is still accessible to students with limited coding experience, the code documentation, contained in `DeveloperDocumentation.ipynb` provides detailed steps on how to launch locally.

For more advanced users and developers interested in further modifications of the package, the Github repository can also be cloned. Full details for use of the package outside of the web app are also available in `DeveloperDocumentation.ipynb`.

Finally, for users interested in learning more about the underlying theory of automatic differentiation, the software is complemented by an accompanying automatic differentiation unit.

Accompanying Automatic Differentiation Unit

This software package is accompanied by a series of learning modules available on Read the Docs to help students understand the theory behind automatic differentiation that is performed and visualized by the package. In the first module, we motivate the need for automatic differentiation, contrast it with numeric and symbolic differentiation, and introduce the basics of forward mode for a single-input, single-output function. In the second module, we expand on the first module to include more of the theory underlying the forward mode, including a consideration of multiple input variables. We also emphasize the computational table and the graph structure in more detail. The third module

introduces the reverse mode of automatic differentiation and connects it to the famous backpropagation algorithm. The fourth module concludes with a series of possible extensions and a discussion of how automatic differentiation might be performed in software. The fourth module has been used to help students focus their final software development project. Each module is accompanied by a series of exercises, where manual exercises are complemented with the `Auto-eD` web application.

Experience of Use

A similar structure of course modules has been used to teach these concepts in the CS107/CS207/AC207 class at the Institute for Applied Computational Science at Harvard since Fall 2018. In Fall 2019, the course introduced a GUI based on portions of this software to help students with the graph visualization, which received positive feedback from students taking the course. This GUI has since been refactored for the web interface, making it more accessible across different operating systems. Students in the Fall 2020 class used web application and responses were positive.

Learning Objectives

Upon completion of this unit, students should be able to: - Explain why automatic differentiation is a valuable computational tool - Decompose a function into a series of elementary operations and write out the associated graph structure - Perform automatic differentiation for functions of single and multiple variables in the forward and reverse mode - Start thinking about how to implement the forward mode of automatic differentiation in software

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